```
from google.colab import files
print("Please upload your 2 files from Downloads: ")
uploaded=files.upload()
```

```
Please upload your 2 files from Downloads:
```

Choose Files 2 files

QVI_transaction_data.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 11979155 bytes, last modified: 28/1/2026 - 100% done

QVI_purchase_behaviour.csv(text/csv) - 2452463 bytes, last modified: 28/1/2026 - 100% done

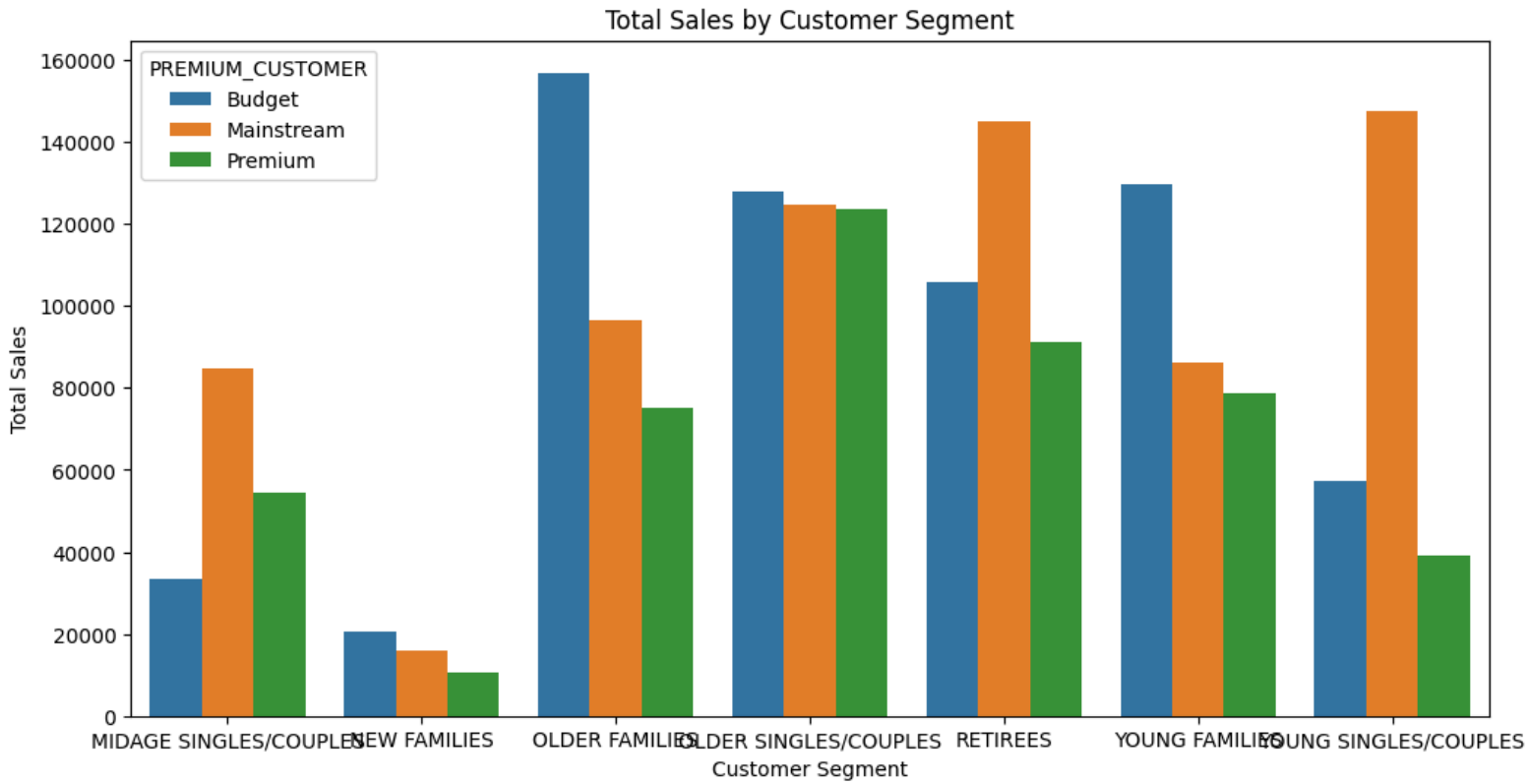
```
Saving QVI_transaction_data.xlsx to QVI_transaction_data.xlsx
Saving QVI_purchase_behaviour.csv to QVI_purchase_behaviour (1).csv
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
print("Starting the full process...")
print("Loading Files...")
transaction_data=pd.read_excel('QVI_transaction_data.xlsx')
customer_data=pd.read_csv('QVI_purchase_behaviour.csv')
print("Cleaning Data...")
transaction_data['DATE']=pd.to_datetime(transaction_data['DATE'],unit='D',origin='1899-12-30')
transaction_data=transaction_data[~transaction_data['PROD_NAME'].str.contains("salsa",case=False)]
transaction_data=transaction_data[transaction_data['PROD_QTY']<200]
transaction_data['PACK_SIZE']=transaction_data['PROD_NAME'].str.extract('(\d+)').astype(float)
transaction_data['BRAND']=transaction_data['PROD_NAME'].str.split().str[0]
brand_replacements={'RRD': 'Red','Smith': 'Smiths','Dorito': 'Doritos','Grnwves': 'Grain','Infzns': 'Infuzions','WW': 'Woolworths','Snbts': 'Sunbites'}
transaction_data['BRAND']=transaction_data['BRAND'].replace(brand_replacements)
print('Merging files...')
merged_data=pd.merge(transaction_data,customer_data,on='LYLTY_CARD_NBR',how='left')
merged_data['TOT_SALES']=pd.to_numeric(merged_data['TOT_SALES'])
print("-"*30)
print("Transaction Rows: ",transaction_data.shape[0])
print("Merged Rows: ", merged_data.shape[0])
print("Missing Details: ",merged_data['LIFESTAGE'].isnull().sum())
print("-"*30)
print("SUCCESS! 'merged_data' is ready for analysis")
```

```
<>:12: SyntaxWarning: invalid escape sequence '\d'
<>:12: SyntaxWarning: invalid escape sequence '\d'
/tmp/ipython-input-2574113869.py:12: SyntaxWarning: invalid escape sequence '\d'
  transaction_data['PACK_SIZE']=transaction_data['PROD_NAME'].str.extract('(\d+)').astype(float)
Starting the full process...
Loading Files...
Cleaning Data...
Merging files...
------------------------------
Transaction Rows:  246740
Merged Rows:  246740
Missing Details:  0
------------------------------
```

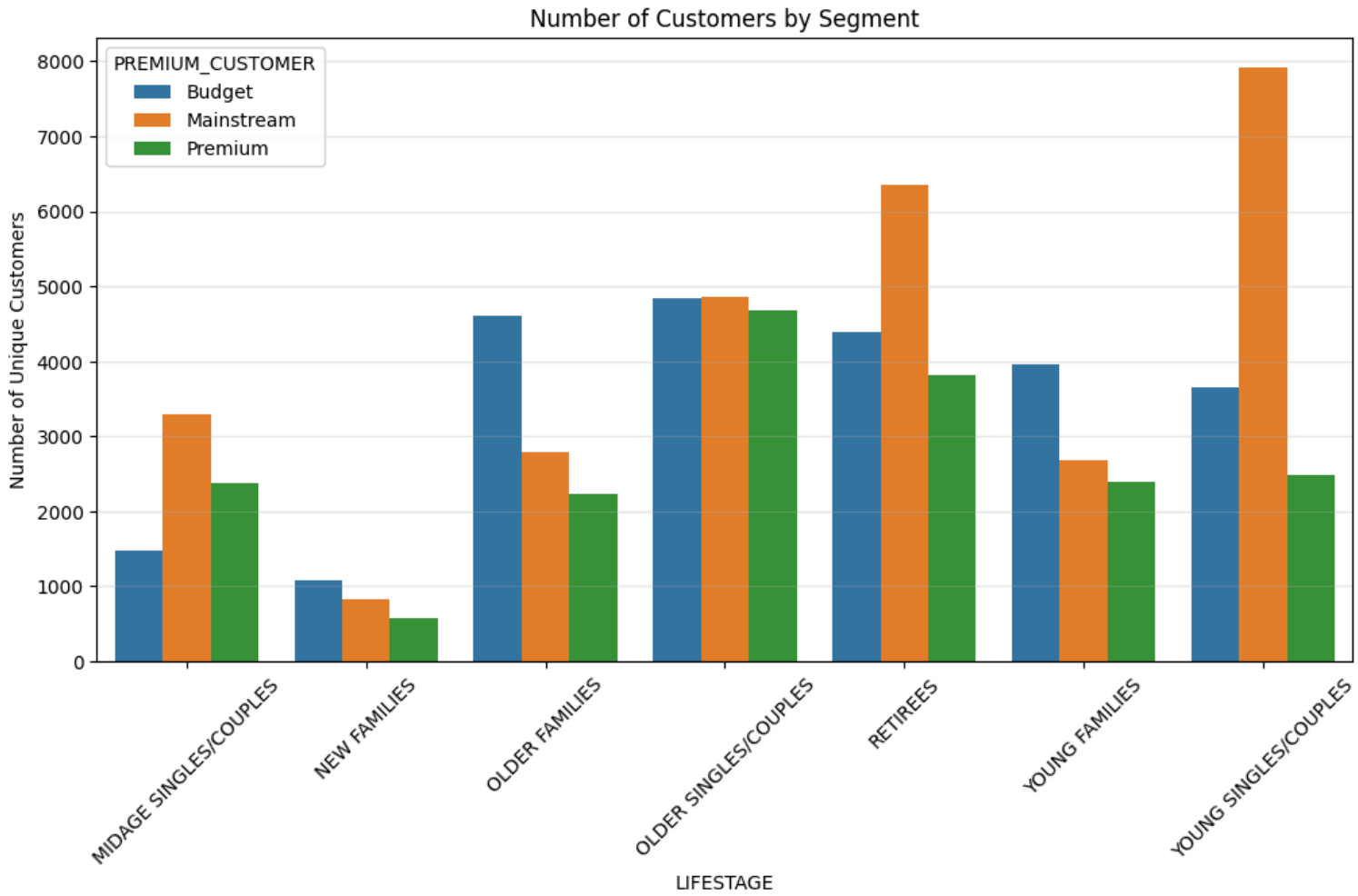
```
SUCCESS! 'merged_data' is ready for analysis
```

```
import matplotlib.pyplot as plt
import seaborn as sns
sales_breakdown=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['TOT_SALES'].sum().reset_index()
plt.figure(figsize=(12,6))
sns.barplot(x='LIFESTAGE',y='TOT_SALES', hue='PREMIUM_CUSTOMER',data=sales_breakdown)
plt.title('Total Sales by Customer Segment')
plt.ylabel('Total Sales')
plt.xlabel('Customer Segment')
plt.show()
```

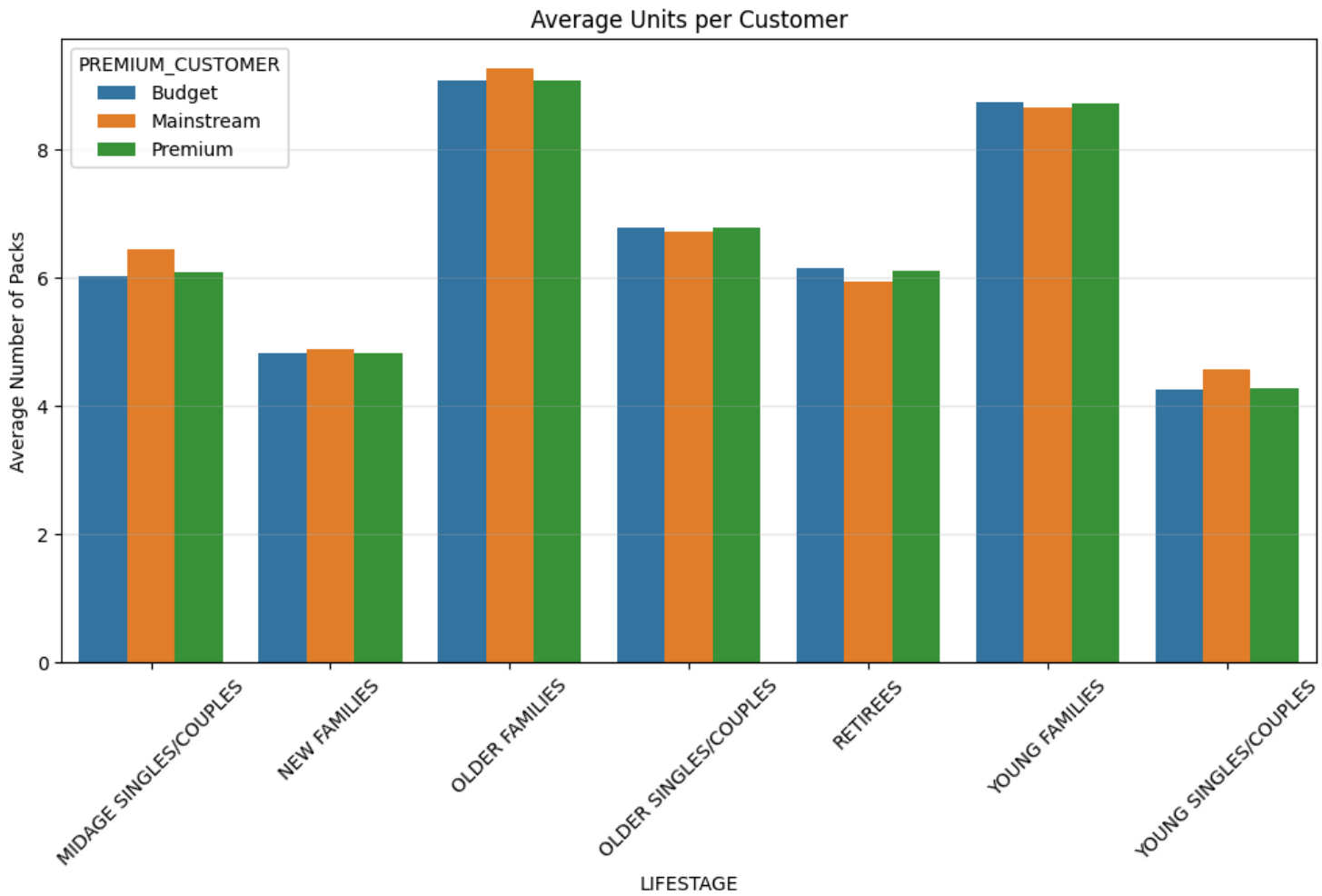



```
customer_counts=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['LYLTY_CARD_NBR'].nunique().reset_index()
plt.figure(figsize=(12,6))
sns.barplot(x='LIFESTAGE', y='LYLTY_CARD_NBR', hue='PREMIUM_CUSTOMER',data=customer_counts)
plt.title('Number of Customers by Segment')
plt.ylabel('Number of Unique Customers')
plt.xticks(rotation=45)
plt.grid(axis='y',alpha=0.3)
```

```
plt.show()
```




```
quantity_per_group=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['PROD_QTY'].sum()
customers_per_group=merged_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER'])['LYLTY_CARD_NBR'].nunique()
avg_units=(quantity_per_group/customers_per_group).reset_index(name='AVG_UNITS')
plt.figure(figsize=(12,6))
sns.barplot(x='LIFESTAGE', y='AVG_UNITS', hue='PREMIUM_CUSTOMER',data=avg_units)
plt.title('Average Units per Customer')
plt.ylabel('Average Number of Packs')
plt.xticks(rotation=45)
plt.grid(axis='y',alpha=0.3)
plt.show()
```



Summary of Findings

Top Spenders: "Older Familes(Budget)" and "Young Singles/Couples(MainStream)" generate the highest Sales.

Drivers of Sale: (1)Young Singles have highest sales because there is a large population of them.

(2)Older Familes have more sales because they buy more packets per customer.

Recomendation: Marketing Strategy should approach these two distict behaviours differently.